

Project Proposal: Global Energy Transition Trends

This project looks at how countries are shifting from fossil fuels to renewables. Using decades of energy consumption data, I want to find out which countries are changing fastest, what drives the pace of that change, and whether renewables are actually replacing fossil fuels or just meeting new demand.

The data comes from the Our World in Data energy dataset, hosted on Kaggle at <https://www.kaggle.com/datasets/pralabhpoudel/world-energy-consumption>. It has one row per country per year, with over 100 columns covering energy consumption by source (coal, oil, gas, nuclear, solar, wind, hydro, biofuel), electricity mix, per-capita figures, GDP, and population. Sources include the Energy Institute, the U.S. EIA, and Ember. The data is clean and well-documented with a full codebook.

Questions I plan to address:

Which countries increased their renewable share the fastest over the past 20 years?

Is there a GDP-per-capita threshold where countries start adopting renewables?

How do the energy mixes of similarly sized economies compare (e.g. Germany vs Japan)?

Are renewables replacing fossil fuels or just covering additional demand?

The plan is four weeks. Week 1: download, clean, and prepare the data. Week 2: exploratory analysis and visualizations. Week 3: comparative and correlation analysis across countries. Week 4: write up findings and finalize the report.